Parity-weighted rank deficits in the memory core
Exact signed moments and a double-scaled Gram matrix

Alan Day Chengheng*

Local research note — 6 November 2026

Abstract

The output-rank deficit in the preceding memory formula is governed by a symmetric-product selection rule. We derive its first two parity-weighted moments from the existing Racah difference equation, without assuming a complete Poisson law. This gives the limiting Gram matrix of the energy-square, centered input-deficit, and output-deficit operators. The output direction remains nonzero after the energy-square projection and is orthogonal to the input direction; its comparison with the force operator A remains open.

1 The finite half-filled rank problem

Take $N = 2m$, standard CAR, and the maximally mixed charge-m state with normalized trace tr_m. Set $D = \binom{N}{m}$, $d_r = \binom{N}{r}^2 - \binom{N}{r-1}^2$, and $x_r = r(N+1-r)$. The neutral Casimir has eigenvalue $2x_r$ on rank r. For independent isotropic Hermitian operators of ranks a, b with $\mathbb{E}U_a^2 = \mathbb{E}V_b^2 = I$, use the previously derived positive law

$$\pi_{ab}(r) = \mathbb{E}\|P_r(U_aV_b)\|_m^2, \qquad \sum_r \pi_{ab}(r) = 1, \qquad r \le a+b. \tag{1}$$

The pure-rank covariance eigenvalues are

$$\chi_a(b) = {}_4F_3\left[\begin{matrix}-a, a-N-1, -b, b-N-1 \\ -m, -m, -N-1\end{matrix}; 1\right]. \tag{2}$$

They are the known Racah specialization [1]. Let

$$S = a+b-r, \qquad X = x_{a+b} - x_r, \qquad f = \chi_a(b). \tag{3}$$

Then the exact signed moments, with expectation under π_{ab}, are

$$\mathbb{E}_\pi(-1)^S = f, \tag{4}$$

$$\mathbb{E}_\pi[(-1)^S X] = -2abf, \tag{5}$$

$$\mathbb{E}_\pi[(-1)^S X^2] = (4a^2b^2 - 2ax_b)f + a^2(m+1-a)[\chi_{a-1}(b) - f]. \tag{6}$$

For $a = 0$ the last term is defined as zero. The formulas include all output ranks and keep the sign; signed moments need not be positive.

*AI research assistant (OpenAI).

2 A proof from the Racah transform

At half filling the linear particle-hole anti-automorphism $\tau(X) = \star X^{\mathsf{T}} \star^{-1}$ acts on rank r by $(-1)^r$. Let T_μ be a full Hermitian matrix basis orthonormal for normalized trace. Completeness gives

$$\sum_\mu (T_\mu)_{ij}(T_\mu)_{kl} = D\delta_{il}\delta_{jk}, \qquad \sum_\mu \tau(T_\mu) X T_\mu = (-1)^m D\tau(X). \tag{7}$$

The second identity uses $\star^{\mathsf{T}} = (-1)^m \star$. Equivalently, for normalized pure-rank covariance maps Φ_r,

$$\sum_r (-1)^r d_r \Phi_r = (-1)^m D\tau, \qquad \sum_r (-1)^r d_r \chi_r(j) = (-1)^{m+j} D. \tag{8}$$

Specialize the existing difference equation [1, Eq. (1.2.5)] with $\alpha = \beta = -m-1$, $\gamma = -N-2$, $\delta = 0$. Define

$$(Lf)(j) = B_j[f(j) - f(j+1)] + D_j[f(j) - f(j-1)], \tag{9}$$

$$B_j = \frac{(j-m)(j-N-1)^2}{2(2j-N-1)}, \qquad D_j = \frac{j^2(j-m-1)}{2(2j-N-1)}. \tag{10}$$

The endpoints have $B_m = D_0 = 0$, and

$$L\chi_a = x_a\chi_a, \quad 2(B_j + D_j) = K - x_j, \quad Lx_j = N(x_j - K/2), \qquad K = \frac{N(N+1)}{2}. \tag{11}$$

Writing $\varepsilon_j = (-1)^j$, conjugation gives

$$\varepsilon L \varepsilon = K - x - L. \tag{12}$$

Insert Eq. (8) into the established Racah triple sum for π. Multiplication by x_r becomes L on the transformed index. If X denotes multiplication by x_a and $\sigma_k = \mathbb{E}_\pi[(-1)^S x_r^k]$, the first two operations give

$$\sigma_1 = (\mathsf{X} + L_a)\chi_a(b) = (x_a + x_b)f, \tag{13}$$

$$\sigma_2 = \left[(\mathsf{X} + L_a)^2 + N(K/2 - \mathsf{X} - L_a)\right]\chi_a(b). \tag{14}$$

Use the recurrence once more to eliminate $f(a+1)$. In $\mathbb{E}(-1)^S X^2$, all order-N^3 terms cancel, leaving Eq. (6). This stable expression needs only $\chi_a(b)$ and $\chi_{a-1}(b)$, rather than a full output-rank sum.

3 Signed and even-weighted double-scaled moments

Let $p^2/N \to \lambda > 0$, $a/p \to \alpha$, $b/p \to \beta$, and write $\theta = 2\lambda\alpha\beta$. The terminating Racah series is dominated uniformly for $a, b \le p$ by $K_\lambda^k/k!$. Each fixed summand tends to $(-4\lambda\alpha\beta)^k/k!$. Hence

$$\chi_a(b) \longrightarrow e^{-2\theta}, \qquad \chi_{a-1}(b) - \chi_a(b) \longrightarrow 0. \tag{15}$$

The prefactor $a^2(m+1-a)/N^2$ in Eq. (6) is bounded. The three exact formulas consequently imply

$$\left(\mathbb{E}(-1)^S,\ \mathbb{E}(-1)^S\frac{X}{N},\ \mathbb{E}(-1)^S\frac{X^2}{N^2}\right) \longrightarrow e^{-2\theta}(1, -\theta, \theta^2 - \theta). \tag{16}$$

For $a + b \le 2p$, the exact identity

$$X = S[N + 1 - 2(a + b) + S] \tag{17}$$

gives $X/N = S[1 + O(p/N)]$ uniformly. The previous positive moment bounds therefore transfer the first two signed limits from X/N to S. With the previous unsigned moments, this yields

$$\mathbb{E}[1 + (-1)^S] \to 1 + e^{-2\theta}, \tag{18}$$

$$\mathbb{E}[S(1 + (-1)^S)] \to \theta(1 - e^{-2\theta}), \tag{19}$$

$$\mathbb{E}[S^2(1 + (-1)^S)] \to \theta^2(1 + e^{-2\theta}) + \theta(1 - e^{-2\theta}). \tag{20}$$

These determine the Gram matrix needed below. They do not establish a complete Poisson distribution.

The actual symmetric product selects precisely these even channels:

$$P_r(V_bU_a) = (-1)^S P_r(U_aV_b), \tag{21}$$

$$\mathbb{E}\|P_r\{U_a, V_b\}\|_m^2 = 4\mathbf{1}_{\{S \text{ even}\}}\pi_{ab}(r). \tag{22}$$

The first equation follows by applying τ to the ordered product. It is an identity for the multiplication map, not an assertion that all product multiplicities are absent. In particular, the $S = 1$ channel is removed. Ignoring this selection rule would give the wrong small-λ strength for the output-deficit operator.

4 The same-coupling quadratic Gram matrix

Return to the full overlapping complex Gaussian p-body Hamiltonian $H = \sum_a h_a$, $\mathbb{E}J_{IJ}J_{KL} = w\delta_{IL}\delta_{JK}$, $\hbar = 1$, at exact half filling. Write $\mathbb{E}h_a^2 = C_a I$, $C = \sum_a C_a$, $\omega_a = C_a/C$, and $\bar{s}_N = \sum_a (p-a)\omega_a$. Define

$$B_N = W_2 H^2 = H^2 - CI, \tag{23}$$

$$L_N = W_2\{H, D_N - \bar{s}_N H\}, \qquad D_N = \sum_a (p-a)h_a, \tag{24}$$

$$\mathcal{S}_N = \sum_{a\le b,r} (a+b-r) W_2 Z^+_{ab;r}. \tag{25}$$

Here W_2 means ordinary Gaussian coupling chaos, and Z^+ uses an anticommutator for $a < b$ and a square for $a = b$.

For any two real symmetric rank filters f, g, orthogonality of input pairs and output ranks, followed by Gaussian decoupling on the diagonal pair, gives the exact identity

$$\left\langle \sum f W_2 Z^+, \sum g W_2 Z^+ \right\rangle = C^2 \sum_{a,b} \omega_a\omega_b \sum_r [1 + (-1)^S]\pi_{ab}(r) f_{ab;r} g_{ab;r}. \tag{26}$$

All scalar products are annealed normalized HS products. Use $f = (1, u + v - 2\bar{s}_N, S)$ for $(B_N, L_N, \mathcal{S}_N)$. The known input deficits tend to independent Poisson variables of mean λ in this weighted sum, so $\bar{s}_N \to \lambda$. For bounded input deficits Eq. (20) has $\theta = 2\lambda$ and $q = e^{-4\lambda}$. The input factorial tail and output third-moment bound justify convergence of every Gram entry, including the mixed entries. Thus

$$\frac{G}{C^2} \longrightarrow \begin{pmatrix} 1+q & 0 & \theta(1-q) \\ 0 & 2\lambda(1+q) & 0 \\ \theta(1-q) & 0 & \theta^2(1+q) + \theta(1-q) \end{pmatrix}. \tag{27}$$

Subtract the energy-square projection, $\mathcal{S}_\perp = \mathcal{S}_N - \mu B_N$, $\mu = \theta(1-q)/(1+q)$. In the physical normalization $C \to J_s^2/\lambda$,

$$\|\mathcal{S}_\perp\|_2^2 \longrightarrow J_s^4 \left[\frac{2(1-q)}{\lambda} + \frac{16q}{1+q}\right] > 0, \qquad \langle \mathcal{S}_\perp, L_N\rangle \to 0. \tag{28}$$

After this fixed-λ limit, the norm tends to $16J_s^4$ as $\lambda \to 0$. The old linear-filter bridge relates L_N to $J_H F$: $J_H F - 4\lambda q C L_N \to 0$ in L^2. The product extension uses the Gaussian fourth-moment bound for the linear bridge error. Hence $\langle \mathcal{S}_\perp, J_H F\rangle \to 0$ as well.

5 Meaning, remaining work, and evidence

The preceding memory-core result contains an output-deficit term $8\lambda C q^2 \mathcal{S}_N$. We now know its equal-time norm and its cross terms with H^2 and the input-deficit direction. The nonzero projected norm shows that the output filter is not exhausted by these two directions in the limiting HS space.

The cross term with the force observable $A = \sum_i [H, c_i]^\dagger [H, c_i]$ is not determined by Eq. (27). It involves the coherent lower-charge product. Consequently this calculation does not yet establish independence from the full old span including A, a nonzero dynamical spectral weight, or a new bulk particle. A projection on H^2 alone is not the full projection onto conserved observables.

The next calculation should use the adjacent-charge covariance and reduced-product recoupling to obtain the A cross and weighted tails. These are needed to replace its exact rank-dependent memory coefficient by the fixed-deficit limit. Then the low-chaos pieces of the fourth-chaos descendants can be restored in raw Z_2. The full charge-window extension, finite-memory covariance, and bulk space/mass dictionary remain part of the original objective.

Exact verification. For $N = 2, 4, 6, 8, 10, 12$ at half filling, all allowed input-rank pairs give 417 signed-moment checks through second order. The same run checks 139 complete Racah recurrences and the parity transform. Independent CAR calculations use all symmetric Gaussian coordinate pairs in three systems: 136 pairs for $(4, 1, 2)$ and 666 each for $(4, 2, 2)$ and $(6, 1, 3)$. Their full 3×3 coefficient Grams agree exactly with Eq. (26); these are not Monte Carlo estimates.

Finite-size verification. Twenty-four cases at $\lambda = 0.1, 0.25, 0.5$, $p = 16, 32, 64, 128$, and input deficits $(0, 0), (1, 2)$ evaluate the exact finite moment formulas at 80 and 120 digits. The measured finite variable is X/N, not S. Actual finite-size deviations are retained. At $\lambda = 0.5, p = 128, (u, v) = (1, 2)$, the signed moment deviations are approximately $(0.004323, -0.001067, -0.002125)$. Decimal precision agreement is not interval certification.

The initial exact checker accidentally evaluated a negative integer power of -1 in a zero-probability channel, causing conversion to floating point. Replacing that expression by integer parity restored exact arithmetic; the initial code is preserved. No formula or tolerance was altered to obtain agreement.

References

[1] R. Koekoek and R. F. Swarttouw, *The Askey-scheme of hypergeometric orthogonal polynomials and its q-analogue*, report 98-17 (1998), Sec. 1.2, Eq. (1.2.5), author-hosted report.